

RATIONAL APPROXIMATION WITH DIGIT-RESTRICTED DENOMINATORS

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ABSTRACT. We show the existence of “good” approximations to a real number γ using rationals with denominators formed by digits 0 and 1 in base b . We derive an elementary estimate and enhance this result by managing exponential sums.

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1. INTRODUCTION

¹ Given a set $H \subseteq \mathbb{N}$ of positive natural numbers, we are interested in whether we can obtain that for any $\gamma \in \mathbb{R}$ and $N \in \mathbb{N}$, one has

$$\min_{1 \leq n \leq N, n \in H} \|\gamma n\| \leq f(N),$$

where $\|\cdot\|$ is the distance to the nearest integer function and f is some function decaying to zero, independent of γ . We say that the set H is *Heilbronn* if the above property is satisfied. In [8, p. 35] Heilbronn sets have been defined in a different way, they say that H is Heilbronn if for all $\gamma \in \mathbb{R}$ and $\varepsilon > 0$ there exists $h(\gamma, \varepsilon) \in H$ so that $\|\gamma h(\gamma, \varepsilon)\| < \varepsilon$. Due

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to a compactness argument these two listed definitions are equivalent [8, p. 35].

A simple example of a Heilbronn set is $\mathbb{N}$, since for any $\gamma \in \mathbb{R}$ we can exploit the Dirichlet approximation Theorem to write

$$\min_{1 \leq n \leq N} \|\gamma n\| \leq \frac{1}{N}.$$

If a set H is discovered to be Heilbronn, along with a suitable decay function f then for any $N \in \mathbb{N}$ and $\gamma \in \mathbb{R}$, there exist a and b in the set $\{1, \dots, N\} \cap H$ so that

$$\left| \gamma - \frac{a}{b} \right| \leq \frac{1}{bf(N)}.$$

With this interpretation of a Heilbronn set, we discover how well rationals with denominators in H approximate real numbers, “how well” being dependent on the quality of the decay function f .

One need not be content after possibly discovering that a certain set H is Heilbronn; discovering a stronger decay function or showing that the function is optimal (with notions of optimality) can be one avenue for the next research stage. In the case of square integers, Heilbronn demonstrated that for any $\gamma \in \mathbb{R}$, one has

$$(1.1) \quad \min_{1 \leq n \leq N} \|\gamma n^2\| \leq C_\theta N^{-\theta},$$

for all $\theta < \frac{1}{2}$, where C_θ is some real number depending only on θ [6]. Later, this proposition was improved to $\theta < \frac{4}{7}$ by Zaharescu [10].

Expanding beyond rational approximations with square or unrestricted denominators, if Q is a polynomial with integer coefficients, positive for natural numbers, with $Q(0) = 0$, then the set of values obtained by applying Q to natural numbers, denoted as $Q(\mathbb{N})$, forms a Heilbronn set [1]. To add to this growing collection of literature, if $b \in \{2, 3, \dots\}$, we prove that the set $\mathfrak{D}_b$, consisting of positive integers made of digits 0 and 1 in base b representation, is a Heilbronn set.

$$\mathfrak{D}_b := \left\{ \sum_{d=0}^L a_d b^d : L \in \mathbb{N} \cup \{0\}, a_d \in \{0, 1\} \right\}.$$

Through elementary means, it is possible to prove that for any $\gamma \in \mathbb{R}$ we have

$$(1.2) \quad \min_{1 \leq n \leq N, n \in \mathfrak{D}_b} \|\gamma n\| \leq \frac{A_b}{\log N},$$

for some effective real constant A_b , depending only on b . Hence, we use (1.2) as a benchmark for further improvements. We enhance the decay function with the help of exponential sums as follows.

Theorem 1.1. *For any $\gamma \in \mathbb{R}$, we have*

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b} \|\gamma n\| \leq \frac{C_b}{(\log N)^2},$$

for some effective real constant C_b , depending only on b .

When u/v is a reduced fraction, the theorem above provides insights into the distribution of $un \bmod v$ as n varies over $\mathfrak{D}_b$. In this context, Theorem 1.1 serves as a complement to Theorem 1 from [5].

Reverting attention to the case of squares, researchers have conjectured that (1.1) holds for $\theta < 1$, see [6, 10]. Inspired by the conjecture, we ask whether for all $\theta < 1$ there exists a constant $C_\theta > 0$, independent of $\gamma \in \mathbb{R}$, so that

$$\min_{1 \leq n \leq N} \|\gamma b_n\| \leq C_\theta N^{-\theta},$$

where $b_1 < b_2 < \dots$ represent all the elements of $\mathfrak{D}_b$. This is not true.

Theorem 1.2. *For any $N \in \mathbb{N}$, there exists $\gamma_N \in \mathbb{R}$, so that*

$$\min_{1 \leq n \leq N} \|\gamma_N b_n\| \geq \frac{1}{b^4} N^{-\frac{\log_2 b}{b-1}}.$$

We can also explore the rational approximation properties of other sets with “missing digits”. It can be observed that sets $S \subseteq \mathbb{Z}$ that exclude the digit 0 in their base b representation do not form a Heilbronn set. This is because for any $w \in S$, we have $\|w \frac{1}{b}\| \geq \frac{1}{b}$.

For further reading on the properties of integers with digit restrictions, see [3, 4].

2. DEFINITIONS AND CONVENTIONS

We list the following:

- $\mathbb{N}$ denotes the set $\{1, 2, \dots\}$.
- γ is a real parameter.
- $\lceil \gamma \rceil$ denotes the smallest integer greater than or equal to γ .
- $\lfloor \gamma \rfloor$ denotes the largest integer less than or equal to γ .
- $\{\gamma\}$ denotes the fractional part of gamma, defined as $\{\gamma\} = \gamma - \lfloor \gamma \rfloor$.
- For $x \in \mathbb{R}$ it is designated that $e(x) := e^{2\pi i x}$.
- $\ll$ is the Vinogradov symbol. That is, for any quantities U and V , we have the following equivalent definitions (where U and V are defined):

$$U \ll V \iff U = O(V) \iff |U| \leq c|V|,$$

for some constant $c > 0$, which throughout the paper is allowed to depend on the parameter $b \in \{2, 3, \dots\}$, but not on the parameters $N \in \mathbb{N}$ and $\gamma \in \mathbb{R}$.

3. SOME ELEMENTARY OBSERVATIONS

Let $t(b, N) = \lfloor \log_b(N(b-1) + 1) - 1 \rfloor$ so that $t(b, N)$ is the largest integer t satisfying $1 + b + \dots + b^t \leq N$, and put

$$P_{N,b} = \left\{ s : s = \sum_{j=0}^d b^j, d \in \{0, \dots, t(b, N)\} \right\}.$$

For each $\gamma \in \mathbb{R}$ either there exists $u \in P_{N,b}$ with $\|u\gamma\| \leq 1/(t(b, N) + 1)$ and (1.2) follows, or by the pigeonhole principle, there exists $u, v \in P_{N,b}$ with $u \neq v$ and $h \in \{0, \dots, t\}$ so that

$$\frac{h}{t(b, N) + 1} \leq \{u\gamma\} \leq \{v\gamma\} \leq \frac{h+1}{t(b, N) + 1},$$

and thus $\|v - u\|\gamma\| \leq 1/(t(b, N) + 1)$, where it should be noted that

$$|v - u| \in \mathfrak{D}_b \cap [1, N],$$

and (1.2) follows again.

Before employing exponential sums to strengthen (1.2), it is prudent to inquire whether a comprehensive application of the pigeonhole principle and the method of differencing could yield a result of comparable or greater strength than Theorem 1.1. After creating sufficient technology, we will see that the answer is no.

Definition 3.1. For a set $S \subseteq \mathbb{N}$, we define

$$\begin{aligned} D^+(S) &:= \{y - x : y, x \in S, y > x\} \\ M_1^+(S) &:= \max\{\#J : J \subseteq \mathbb{Z}, D^+(J) \subseteq S\} \\ M_2^+(S) &:= \max\{\#J : J \subseteq S, D^+(J) \subseteq S\}. \end{aligned}$$

One observes that by a similar argument used to show (1.2) (using the pigeonhole principle), we can show that for any non-empty set $S \subseteq \mathbb{N}$ and $\gamma \in \mathbb{R}$, one has

$$\min_{n \in S} \|\gamma n\| \leq \frac{1}{\max\{M_1^+(S) - 1, M_2^+(S) + 1\}}.$$

By the trivial inequality $M_2^+(S) \leq M_1^+(S)$, if we manage to see that $M_1^+(\mathfrak{D}_b \cap \{1, \dots, N\}) \ll \log N$ whenever $b \geq 3$ we could convince ourselves that such elementary methods are not enough to strengthen (1.2) sufficiently.

Lemma 3.2. *For $b \geq 3$ we have*

$$M_1^+(\mathfrak{D}_b \cap [1, N]) \ll \log N.$$

Proof. Suppose that $A(b, N) \subseteq \mathbb{Z}$ satisfies $\#A(b, N) = M_1^+(\mathfrak{D}_b \cap [1, N])$ and $D^+(A(b, N)) \subseteq \mathfrak{D}_b \cap [1, N]$. We may list the elements of $A(b, N)$ in an increasing order as $v_1 < \dots < v_z$ and put

$$P = ((v_{j+1} - v_j))_{j=1}^{z-1},$$

a sequence of $z - 1$ differences.

We know that each element $v_{j+1} - v_j$ of P is if the form

$$\sum_{d=0}^{\lfloor \log_b N \rfloor} \xi_{d,j} b^d,$$

where $\xi_{d,j} \in \{0, 1\}$ and $\xi_{d,j} \neq 0$ for some $d \in \{0, \dots, \lfloor \log_b N \rfloor\}$.

We claim that for $1 \leq j_1 < j_2 \leq z - 1$ not both ξ_{d,j_1} and ξ_{d,j_2} are 1; indeed for the sake of contradiction suppose that there exists the smallest value $a \in \{1, \dots, z - 2\}$, for which there exists $h_a \in \mathbb{N}$ with $1 \leq h_a \leq z - 1 - a$, $Y(a, h_a) \in \{0, \dots, \lfloor \log_b(N) \rfloor\}$ and

$$\xi_{Y(a, h_a), h_a} = \xi_{Y(a, h_a), h_a + a} = 1.$$

Then

$$v_{h_a + a} - v_{h_a} = \sum_{t=0}^{a-1} v_{h_a + t + 1} - v_{h_a + t} = \sum_{d=0}^{\lfloor \log_b(N) \rfloor} \rho_d b^d,$$

where $\rho_d \in \{0, 1, 2\}$ and $\rho_{Y(a, h_a)} = 2$, implying $v_{h_a + a} - v_{h_a} \notin \mathfrak{D}_b$, which is impossible.

For each $j \in \{1, \dots, z - 1\}$ put

$$K_j := \{d : \xi_{d,j} = 1\},$$

so that $\#K_j \geq 1$,

$$\bigcup_{j=1}^{z-1} K_j \subseteq \{0, \dots, \lfloor \log_b N \rfloor\}$$

and $K_{j_1} \cap K_{j_2} = \emptyset$ if $j_1 \neq j_2$. These constraints imply that

$$z - 1 \leq \lfloor \log_b N \rfloor + 1,$$

and therefore

$$M_1^+(\mathfrak{D}_b \cap [1, N]) \leq \lfloor \log_b N \rfloor + 2.$$

□

In our path towards the proof of Theorem 1.1, we employ the set $\mathfrak{D}_b^*$ defined as

$$\mathfrak{D}_b^* := \mathfrak{D}_b \cup \{b^d - b^c : d, c \in \mathbb{N} \cup \{0\}, d > c\},$$

and obtain a “weaker” version of Theorem 1.1.

Lemma 3.3. *For all $\gamma \in \mathbb{R}$, we have*

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b^*} \|\gamma n\| \ll \frac{1}{(\log N)^2},$$

where the implied constant is effective and depends only on b .

In fact, Lemma 3.3 is equally as strong as Theorem 1.1 due to the following assertion:

Lemma 3.4. *If for all $\gamma \in \mathbb{R}$ there is some positive function $f : \mathbb{N} \rightarrow \mathbb{R}$ so that*

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b^*} \|\gamma n\| \leq f(N),$$

then we have

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b} \|\gamma n\| \leq (b-1)f(N).$$

Proof. Let $\gamma^* = \frac{\gamma}{b-1}$. Given the assumption

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b^*} \|\gamma^* n\| \leq f(N),$$

we have two cases to consider:

- If there exists $x \in \mathfrak{D}_b$ such that $1 \leq x \leq N$ and $\|x\gamma^*\| \leq f(N)$, then

$$\|x\gamma\| \leq (b-1)f(N).$$

- If there exist positive integers d and c with $d > c$, and $1 \leq b^d - b^c \leq N$, such that $\|(b^d - b^c)\gamma^*\| \leq f(N)$, then $\|\frac{b^d - b^c}{b-1}\gamma\| \leq f(N)$, where $\frac{b^d - b^c}{b-1} \in \mathfrak{D}_b$ and $1 \leq \frac{b^d - b^c}{b-1} \leq N$, because $1 \leq b^d - b^c \leq N$ and $b^d - b^c$ is a multiple of $b-1$. $\square$

4. EXPONENTIAL SUMS WITH DIGIT RESTRICTIONS

This section establishes upper bounds for exponential sums involving the parameter $\gamma \in \mathbb{R}$. Our focus is on cases where γ exhibits unfavourable behaviour—defined as situations where, for a given $m \in \mathbb{N}$, $\|\gamma x\|$ exceeds $\frac{1}{2b^m}$ for all x in the set $\mathfrak{D}_b^* \cap [1, N]$. If γ is unfavourable for large values of N (with hidden m), our analysis reveals improvements over the trivial upper bound for these sums. In the next section, such bounds on the exponential sums will be used to prove that γ is “not

unfavourable" when $N \geq J_b b^{m/2}$, where J_b is an effective real constant depending only on b .

For $r \in \mathbb{N} \cup \{0\}$, define

$$\mathfrak{D}_b(r) := \left\{ \sum_{k=0}^r \xi_k b^k : \xi_k \in \{0, 1\} \right\},$$

so that $\mathfrak{D}_b(r)$ has 2^{r+1} elements. Also, let

$$\mathfrak{D}_b^*(r) := \mathfrak{D}_b(r) \cup \{b^d - b^c : c, d \in \{0, \dots, r\}, d > c\}.$$

Lemma 4.1. *For $k \in \mathbb{Z}$, $r \in \mathbb{N} \cup \{0\}$ and $\gamma \in \mathbb{R}$ we have the bound*

$$\left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| \leq 2^{r+1} \prod_{d=0}^r (1 - \pi \|kb^d \gamma\|^2).$$

Proof. We can write

$$\begin{aligned} \left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| &= \left| \prod_{d=0}^r (1 + e(kb^d \gamma)) \right| \\ &= \left| \prod_{d=0}^r (e(kb^d \gamma/2) + e(-kb^d \gamma/2)) \right| \\ &= \left| \prod_{d=0}^r 2 \cos(\pi kb^d \gamma) \right| \\ &= 2^{r+1} \left| \prod_{d=0}^r \cos(\pi kb^d \gamma) \right|. \end{aligned}$$

When $x \in \mathbb{R}$, simple calculus shows that

$$(4.1) \quad |\cos(\pi x)| \leq 1 - \pi \|x\|^2.$$

Using (4.1) we obtain the bound

$$2^{r+1} \left| \prod_{d=0}^r \cos(\pi kb^d \gamma) \right| \leq 2^{r+1} \prod_{d=0}^r (1 - \pi \|kb^d \gamma\|^2). \quad \square$$

Lemma 4.2. *Given t integers $a_1, \dots, a_t$ distinct modulo k for some $k \geq 1$, if $t \geq 3\sqrt{k}$ there exists some $h \geq 1$ so that there exists h distinct elements from $\{a_1, \dots, a_t\}$ that sum to a number that is a multiple of k .*

Proof. We can apply [9, Corollary 3.2.1] to the additive group $\mathbb{Z}/k\mathbb{Z}$. $\square$

Lemma 4.3. *If for a fixed $\beta > 0$, $r \in \mathbb{N} \cup \{0\}$ and $\gamma \in \mathbb{R}$, one has $\|\gamma x\| > \beta$ for all $x \in \mathfrak{D}_b^*(r)$ with $x \neq 0$, then for each $k \in \mathbb{N}$ we have*

$$\#\{d : d \in \{0, \dots, r\}, \|kb^d \gamma\| \leq \beta\} < 3\sqrt{k}.$$

Proof. The assertion is trivially true for $k = 1$; assume $k \geq 2$. Suppose $d_1, \dots, d_g$ are distinct integers in the set $\{0, \dots, r\}$ so that $\|kb^{d_j} \gamma\| \leq \beta$. We can extract the integers $s_1, \dots, s_g$ so that $\|kb^{d_j} \gamma\| = |kb^{d_j} \gamma - s_j|$. Observe that if for $1 \leq u < v \leq g$ and $s_u \equiv s_v \pmod k$ we will have that

$$|(k(b^{d_v} - b^{d_u})\gamma - s_v + s_u)| \leq |kb^{d_v} \gamma - s_v| + |-(kb^{d_u} \gamma - s_u)| \leq 2\beta.$$

Hence,

$$|(b^{d_v} - b^{d_u})\gamma + \frac{-s_v + s_u}{k}| \leq \frac{2\beta}{k} \leq \beta.$$

Thus,

$$\|(b^{d_v} - b^{d_u})\gamma\| \leq \beta,$$

which contradicts our assumption. Therefore the set $\{s_1, \dots, s_g\}$ holds g distinct congruences modulo k .

Now observe that if $g > \frac{k}{2}$ there exists $s_{u_1}, s_{u_2} \in \{s_1, \dots, s_g\}$ satisfying $s_{u_1} + s_{u_2} \equiv 0 \pmod k$ and $u_1 \neq u_2$. With these constraints, one would obtain that

$$|k(b^{d_{u_1}} + b^{d_{u_2}})\gamma - s_{u_1} - s_{u_2}| \leq |kb^{d_{u_1}} \gamma - s_{u_1}| + |kb^{d_{u_2}} \gamma - s_{u_2}| \leq 2\beta.$$

Hence,

$$|(b^{d_{u_1}} + b^{d_{u_2}})\gamma - \frac{s_{u_1} + s_{u_2}}{k}| \leq \frac{2\beta}{k} \leq \beta.$$

Thus,

$$\|(b^{d_{u_1}} + b^{d_{u_2}})\gamma\| \leq \beta,$$

which contradicts our assumption. Therefore the inequality $g \leq \frac{k}{2}$ holds, implying $g < 3\sqrt{k}$ for $2 \leq k \leq 35$. For the remainder of the argument, assume $k \geq 36$.

Note that if $g \geq 3\sqrt{k}$, by Lemma 4.2 there exists $h \in \mathbb{N}$ with

$$1 \leq h \leq [3\sqrt{k}],$$

and an increasing sequence of integers satisfying

$$1 \leq u_1 < \dots < u_h \leq [3\sqrt{k}],$$

so that $s_{u_1} + \dots + s_{u_h} \equiv 0 \pmod{k}$. Hence, we can write

$$|k(b^{d_{u_1}} + \dots + b^{d_{u_h}})\gamma - s_{u_1} - \dots - s_{u_h}| \leq \sum_{j=1}^h |k(b^{d_{u_j}})\gamma - s_{u_j}| \leq h\beta.$$

Thus (with $k \geq 36$),

$$|(b^{d_{u_1}} + \dots + b^{d_{u_h}})\gamma - \frac{s_{u_1} + \dots + s_{u_h}}{k}| \leq \frac{h\beta}{k} \leq \frac{[3\sqrt{k}]\beta}{k} \leq \beta.$$

Therefore

$$\|(b^{d_{u_1}} + \dots + b^{d_{u_h}})\gamma\| \leq \beta.$$

Thus, the inequality $g < 3\sqrt{k}$ must hold for $k \geq 36$ and $2 \leq k \leq 35$. $\square$

For $t \geq 1$, we define the sets

$$G_b(t) := \left\{ y : y \in \mathbb{R}, \frac{1}{2b^t} < \|y\| \leq \frac{1}{2b^{t-1}} \right\}.$$

Lemma 4.4. *If $y \in G_b(t)$ for some $t \geq 2$, then $by \in G_b(t-1)$.*

Proof. Assume $y \in G_b(t)$ for some $t \geq 2$. This assumption implies the existence of an integer c such that

$$\frac{1}{2b^t} < |y - c| \leq \frac{1}{2b^{t-1}}.$$

Therefore, we can deduce

$$\frac{1}{2b^{t-1}} < |by - bc| \leq \frac{1}{2b^{t-2}}.$$

Since $b \geq 2$ and $t \geq 2$, it follows that $\frac{1}{2b^{t-2}} \leq \frac{1}{2}$, which implies $\|by\| = |by - bc|$. Consequently, we obtain the inequality

$$\frac{1}{2b^{t-1}} < \|by\| \leq \frac{1}{2b^{t-2}}. \quad \square$$

Lemma 4.5. *Let m be a natural number, $r \in \mathbb{N} \cup \{0\}$, $k \in \mathbb{N}$, and $\gamma \in \mathbb{R}$ such that $\|\gamma x\| > \frac{1}{2b^m}$ for all $x \in \mathfrak{D}_b^*(r)$, $x \neq 0$. Then we have the following bound:*

$$\left| \sum_{j \in \mathfrak{D}_b(r), j \neq 0} e(kj\gamma) \right| \leq 2^{r+3} \left(1 - \frac{\pi}{4b^2} \right)^{\frac{r-3\sqrt{k}+1}{m}}.$$

Proof. First, using Lemma 4.1, we obtain the bound:

$$\left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| \leq 2^{r+1} \prod_{d=0}^r (1 - \pi \|kb^d\gamma\|^2).$$

Let

$$V_{\gamma,k,r}(m) = \left\{ d : d \in \{0, \dots, r\}, \|kb^d\gamma\| > \frac{1}{2b^m} \right\}.$$

Using the product inequality above, we have the estimate:

$$\left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| \leq 2^{r+1} \prod_{d \in V_{\gamma,k,r}(m)} (1 - \pi \|kb^d\gamma\|^2).$$

For each element v in $V_{\gamma,k,r}(m)$, we have $kb^v\gamma \in G_b(h)$ for some $h \in \{1, \dots, m\}$. Using Lemma 4.4 and induction, we can show that $kb^{v+h-1}\gamma \in G_b(1)$. Let $0 \leq a_1 < \dots < a_{\#V_{\gamma,k,r}(m)} \leq r$ be all the elements of $V_{\gamma,k,r}(m)$ arranged in increasing order.

If $\#V_{\gamma,k,r}(m) \geq m$, then we can consider inequalities

$$a_{mt} < \dots < a_{mt+m-1},$$

for $t = 0, \dots, \lfloor \frac{\#V_{\gamma,k,r}(m)}{m} \rfloor - 1$. For each such t , there is an $h_t \in \{1, \dots, m\}$ so that $kb^{a_{mt}+h_t-1}\gamma \in G_b(1)$, and thus $a_{mt} + h_t - 1 \in V_{\gamma,k,r}(m)$. We have

$$a_{mt} \leq a_{mt} + h_t - 1 \leq a_{mt+m-1},$$

and hence $a_{mt} + h_t - 1 \in \{a_{mt}, \dots, a_{mt+m-1}\}$ as $a_{mt} + h_t - 1 \in V_{\gamma,k,r}(m)$. Therefore, for $t = 0, \dots, \lfloor \frac{\#V_{\gamma,k,r}(m)}{m} \rfloor - 1$, the sets $\{a_{mt}, \dots, a_{mt+m-1}\}$ have an element d_t for which $kb^{d_t}\gamma \in G_b(1)$.

Consequently, there are at least $\lfloor \frac{\#V_{\gamma,k,r}(m)}{m} \rfloor$ elements d in $V_{\gamma,k,r}(m)$ that satisfy $kb^d\gamma \in G_b(1)$. We can now establish an upper bound on:

$$\begin{aligned} \left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| &\leq 2^{r+1} \prod_{d \in V_{\gamma,k,r}(m)} (1 - \pi \|kb^d\gamma\|^2) \\ &\leq 2^{r+1} \prod_{d \in V_{\gamma,k,r}(m), kb^d\gamma \in G_b(1)} (1 - \pi \|kb^d\gamma\|^2) \\ (4.2) \quad &\leq 2^{r+1} \prod_{d \in V_{\gamma,k,r}(m), kb^d\gamma \in G_b(1)} \left(1 - \pi \left(\frac{1}{2b} \right)^2 \right) \\ &\leq 2^{r+1} \left(1 - \frac{\pi}{4b^2} \right)^{\lfloor \frac{\#V_{\gamma,k,r}(m)}{m} \rfloor}. \end{aligned}$$

By inspection, we can verify that the above inequality holds even if $\#V_{\gamma,k,r}(m) < m$. Using Lemma 4.3 and $\beta = \frac{1}{2b^m}$, we observe that

$$\#V_{\gamma,k,r}(m) \geq r + 1 - (3\sqrt{k}) = r - 3\sqrt{k} + 1.$$

Therefore, from (4.2) it follows that

$$\left| \sum_{j \in \mathfrak{D}_b(r)} e(kj\gamma) \right| \leq 2^{r+1} \left(1 - \frac{\pi}{4b^2} \right)^{\lfloor \frac{r-3\sqrt{k}+1}{m} \rfloor}.$$

In particular, we have:

$$\left| \sum_{j \in \mathfrak{D}_b(r), j \neq 0} e(kj\gamma) \right| \leq 2^{r+3} \left(1 - \frac{\pi}{4b^2} \right)^{\frac{r-3\sqrt{k}+1}{m}}.$$

□

5. PROOF OF THEOREM 1.1

To make use of our bounds on exponential sums, we employ a result found in [2, Theorem 2.2].

Lemma 5.1. *Let $x_1, \dots, x_N$ be a real sequence and suppose that M is a positive integer with $\|x_j\| \geq M^{-1}$. Then*

$$\sum_{v=1}^M \left| \sum_{n=1}^N e(vx_n) \right| > \frac{N}{6}.$$

Lemma 5.2. *For all $\gamma \in \mathbb{R}$ and $m \in \mathbb{N} \cup \{0\}$, there exists an effective constant $J_b > 0$ (depending only on b), such that if r is an integer with $r \geq J_b b^{m/2}$ then $\|\gamma y\| \leq \frac{1}{2b^m}$ for some $y \in \mathfrak{D}_b^*(r)$, $y \neq 0$.*

Proof. It suffices to prove the theorem for $m \geq 1$. Let all the elements of $\mathfrak{D}_b$ be arranged as $1 = b_1 < b_2 < \dots$. We can observe that when $r \in \mathbb{N} \cup \{0\}$ we have

$$\mathfrak{D}_b(r) = \{0\} \cup \{b_1, \dots, b_{2^{r+1}-1}\}.$$

Suppose that for all $x \in \mathfrak{D}_b^*(r)$, $x \neq 0$ we have $\|\gamma x\| > \frac{1}{2b^m}$. By Lemma 5.1 we have

$$\sum_{v=1}^{2b^m} \left| \sum_{n=1}^{2^{r+1}-1} e(vb_n x) \right| > \frac{2^{r+1}-1}{6},$$

and thus there exists $h \in \{1, \dots, 2b^m\}$ so that

$$(5.1) \quad \left| \sum_{n=1}^{2^{r+1}-1} e(hb_n x) \right| > \frac{2^r}{12b^m}.$$

By Lemma 4.5 one has

$$(5.2) \quad \left| \sum_{n=1}^{2^{r+1}-1} e(hb_n x) \right| \leq 2^{r+3} \left(1 - \frac{\pi}{4b^2} \right)^{\frac{r-3\sqrt{h}+1}{m}} \\ \leq 2^{r+3} \left(1 - \frac{\pi}{4b^2} \right)^{\frac{r-3\sqrt{2b^m}+1}{m}}.$$

Due to (5.1) and (5.2) it is gathered that

$$(5.3) \quad \frac{2^r}{12b^m} \leq 2^{r+3} \left(1 - \frac{\pi}{4b^2} \right)^{\frac{r-3\sqrt{2b^m}+1}{m}}.$$

It is noted that there exists an effective constant $J_b > 0$ so that when $r \geq J_b b^{m/2}$, inequality (5.3) is violated, this proves the Lemma. $\square$

For a given natural $N \in \mathbb{N}$ we recall that

$$t(b, N) = \lfloor \log_b(N(b-1) + 1) - 1 \rfloor,$$

so that $t(b, N)$ is the largest integer t satisfying $1 + b + \dots + b^t \leq N$. We observe that $\mathfrak{D}_b^*(t(b, N)) \subseteq \{0, \dots, N\}$. Put $m_N = \lfloor 2 \log_b(\frac{t(b, N)}{J_b}) \rfloor$, so that

$$t(b, N) = J_b b^{(2 \log_b(\frac{t(b, N)}{J_b})) / 2} \geq J_b b^{m_N / 2}.$$

By Lemma 5.2 there exists $y \in \mathfrak{D}_b^*(t(b, N))$ with $y \neq 0$ so that

$$\begin{aligned} \|\gamma y\| &\leq \frac{1}{2b^{m_N}} \\ &\leq \frac{b}{2b^{2 \log_b(\frac{t(b, N)}{J_b})}} \\ &= \frac{bJ_b^2}{2t(b, N)^2} \\ &= \frac{bJ_b^2}{2[\log_b(N(b-1) + 1) - 1]^2}. \end{aligned}$$

To put it differently, we can write

$$\min_{1 \leq n \leq N, n \in \mathfrak{D}_b^*} \|\gamma n\| \leq \frac{bJ_b^2}{2[\log_b(N(b-1) + 1) - 1]^2},$$

implying Lemma 3.3. With Lemmas 3.3 and 3.4 we have Theorem 1.1.

6. PROOF OF THEOREM 1.2

For $t \in \mathbb{N}$, let

$$\mathcal{Z}_{b,t} := \left\{ \sum_{d=1}^t b^{u_d} : u_d \in \mathbb{N} \cup \{0\} \right\},$$

so that $\mathcal{Z}_{b,t}$ contains sums of t integer powers of b . Let $1 = b_1 < b_2 < \dots$ be all the elements of $\mathfrak{D}_b$ arranged in an increasing order. It is inspected that with $T \in \mathbb{N}$,

$$(6.1) \quad \{b_1, \dots, b_{2^T-1}\} \subseteq \bigcup_{t=1}^T \mathcal{Z}_{b,t}.$$

We now construct preparatory tools to understand the behaviour of n mod $b^k - 1$ as n ranges through $\mathcal{Z}_{b,t}$ when k is a positive integer.

Lemma 6.1. *Let $k \in \mathbb{N}$, and suppose that $u_0, \dots, u_{k-1}$ are non-negative integers satisfying $u_0 + \dots + u_{k-1} > 0$. There exists integers $v_0, \dots, v_{k-1}$ with $v_j \in \{0, \dots, b-1\}$ so that*

$$\sum_{j=0}^{k-1} u_j b^j \equiv \sum_{j=0}^{k-1} v_j b^j \pmod{b^k - 1}$$

and

$$0 < \sum_{j=0}^{k-1} v_j \leq \sum_{j=0}^{k-1} u_j.$$

Proof. Suppose that n is the smallest natural number such that there exist $u_0, \dots, u_{k-1}$ with $\sum_{j=0}^{k-1} u_j = n$, for which the proposition is false. With this choice of $(u_j)_{j=0}^{k-1}$ there exists some $d \in \{0, \dots, k-1\}$ for which $u_d \geq b$ (and thus $n \geq b$). Let $r_0, \dots, r_{k-1}$ be non-negative integers so that $r_d = u_d - b$,

$$r_{d+1-k \lfloor \frac{d+1}{k} \rfloor} = \left(u_{d+1-k \lfloor \frac{d+1}{k} \rfloor} \right) + 1$$

and $r_t = u_t$ when $t \neq d$ and $t \neq d+1-k \lfloor \frac{d+1}{k} \rfloor$.

We observe that $(r_j)_{j=0}^{k-1}$ is a non-negative integer sequence,

$$\sum_{j=0}^{k-1} u_j b^j \equiv \sum_{j=0}^{k-1} r_j b^j \pmod{b^k - 1},$$

and

$$\sum_{j=0}^{k-1} r_j = \sum_{j=0}^{k-1} u_j - (b-1) = n - b + 1.$$

Thus, n would not be the smallest number for which the proposition is false, yielding a contradiction. $\square$

Let $k, t \in \mathbb{N}$, and put

$$\mathfrak{A}(b, k, t) := \left\{ \sum_{j=0}^{k-1} a_j b^j : a_j \in \{0, \dots, b-1\}, 0 < \sum_{j=0}^{k-1} a_j \leq t \right\}.$$

Lemma 6.2. *Let $t, k \in \mathbb{N}$, if $w \in \mathcal{Z}_{b,t}$ then there exists $c_w \in \mathfrak{A}(b, k, t)$ for which*

$$w \equiv c_w \pmod{(b^k - 1)}.$$

Proof. From the definition of the set $\mathcal{Z}_{b,t}$ there exist non-negative integers $r_1, \dots, r_t$ for which $w = \sum_{d=1}^t b^{r_d}$. By modular arithmetic, we gather that

$$w \equiv \sum_{j=0}^{k-1} u_j b^j \pmod{(b^k - 1)},$$

where $u_0, \dots, u_{k-1}$ are non-negative integers that satisfy $u_0 + \dots + u_{k-1} = t$. By Lemma 6.1 there exist integers $v_0, \dots, v_{k-1}$ with $v_j \in \{0, \dots, b-1\}$, $v_0 + \dots + v_{k-1} \leq t$, and

$$\sum_{j=0}^{k-1} u_j b^j \equiv \sum_{j=0}^{k-1} v_j b^j \pmod{(b^k - 1)}.$$

Now observe that $\sum_{j=0}^{k-1} v_j b^j \in \mathfrak{A}(b, k, t)$. $\square$

Lemma 6.3. *Let $k, t \in \mathbb{N}$ with $k > \frac{t}{(b-1)}$, we have $b^k - 1 \notin \mathfrak{A}(b, k, t)$.*

Proof. Let $h(b, k, t)$ be the largest element of $\mathfrak{A}(b, k, t)$. With the constraints of the proposition we have $h(b, 1, t) = t \leq b-2 < b-1$. Suppose for the sake of induction that the proposition has been verified for values $k = 1, \dots, d$. Given that $d+1 > \frac{t}{b-1}$, if $1 \leq t \leq b-1$, then $h(b, d+1, t) = tb^d < b^{d+1} - 1$. If $t \geq b$ one has

$$h(b, d+1, t) = (b-1)b^d + h(b, d, t - (b-1)),$$

where $d > \frac{t-(b-1)}{b-1}$, so that

$$\begin{aligned} h(b, d+1, t) &\leq (b-1)b^d + b^d - 2 \\ &= b^{d+1} - 2. \end{aligned}$$

$\square$

With Lemmas 6.2 and 6.3 we have the following statement.

Corollary 6.4. *Let $k, t \in \mathbb{N}$ with $k > \frac{t}{(b-1)}$, then $\mathcal{Z}_{b,t}$ does not contain multiples of $b^k - 1$.*

Let $T = \lceil \log_2(N+1) \rceil$ so that

$$\{b_1, \dots, b_N\} \subseteq \{b_1, \dots, b_{2^T-1}\}.$$

By the subset relation (6.1) and Corollary 6.4, $\{b_1, \dots, b_{2^T-1}\}$ does not contain multiples of $b^k - 1$ when $k = \lceil \frac{T}{b-1} \rceil + 1$. In particular it is verified that with $\gamma_N = \frac{1}{b^k-1}$, $\|\gamma_N w\| \geq \frac{1}{b^k-1}$ for all $w \in \{b_1, \dots, b_N\}$. Observe that

$$\frac{1}{b^k - 1} \geq \frac{1}{b^{\frac{T}{b-1}+2} - 1} \geq \frac{1}{b^{\frac{T}{b-1}+2}} \geq \frac{1}{b^{\frac{\log_2(N+1)}{b-1}+3}} \geq \frac{1}{b^{\frac{\log_2(N)}{b-1}+4}} = \frac{1}{b^4} N^{-\frac{\log_2 b}{b-1}}.$$

7. CONJECTURES

It is possible that the decay function in Theorem 1.1 is not optimal. Inspired by the form of the decay function presented in (1.1), employed to establish that the set of positive squares is a Heilbronn set, we pose the following question.

Conjecture 7.1. *There exists a constant $c > 0$ so that for all $\gamma \in \mathbb{R}$ we have*

$$\min_{1 \leq n \leq N, n \in \mathcal{D}_b} \|\gamma n\| \ll N^{-c}$$

where the implied constant is independent of γ .

With Theorem 1.2 and a density argument we know that $c \leq \frac{1}{b-1}$. Next, once again turning our attention to the literature on approximating real numbers with rationals having square denominators, we present a further inquiry.

Conjecture 7.2. *The set $\mathcal{D}_{b,2}$, comprising the squares of elements from $\mathcal{D}_b$, formally defined as*

$$\mathcal{D}_{b,2} := \{s^2 : s \in \mathcal{D}_b\},$$

is a Heilbronn set.

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